

Ruoxiao (Roxanna) Zhang

226-978-2694 | r583zhan@uwaterloo.ca | [linkedin.com/in/ruoxiao-zhang](https://www.linkedin.com/in/ruoxiao-zhang) | github.com/ruoxiao-zhang

SUMMARY OF QUALIFICATIONS

- Highly motivated Candidate for Bachelor of Mathematics, demonstrating a strong academic upward trajectory with a Term 1B GPA of 3.96/4.0 and an intended focus on Applied Mathematics with Scientific Computing and Scientific Machine Learning.
- Front-end developer with hands-on experience building intuitive user interfaces that translate complex algorithmic data and APIs into actionable user insights.
- Proven leadership and communication skills, actively partnering with university administration to mentor incoming students and manage official digital communities.
- Certified in Bloomberg Market Concepts, demonstrating a solid understanding of financial markets, economic indicators, and data analysis.
- Proficient programmer with hands-on experience in Python, R, C++, and Java, highlighted by a Gold Medal in the USA Computing Olympiad.

SKILLS, CERTIFICATIONS & PUBLICATIONS

Programming & Tools: Python (Data Analysis/Automation), R (Statistical Computing), C++, Java, Figma, 3D Modeling

Certifications: Bloomberg Market Concepts (BMC), USA Computing Olympiad Gold Medal

Publication: Sole Author, "*The Impact of Artificial Intelligence Represented by ChatGPT on UI Design*", Proceedings of the 2nd International Conference on Software Engineering and Machine Learning, June 2024

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for Bachelor of Mathematics, Honours Mathematics (Co-op)

Sept. 2025 – Present

- **Academic Standing:** Term 1B GPA: 3.96/4.0 | Term 1A Average: 85.6%
- **Relevant Courses:** Intro Computer Science 2 (90%), Intro Macroeconomics (92%), Algebra Honours (92%), Calculus 2 Honours (89%)

Chongqing Nankai Secondary School

Chongqing, China

High School Diploma

Sept. 2022 – July 2025

- **Relevant Courses:** AP Computer Science A, AP Calculus BC, AP Statistics, AP Macroeconomics

RELEVANT EXPERIENCE & CORE PROJECTS

Waterloo Workload Calculator | *Front-End Developer*

Jan. 2026 – Apr. 2026

- Architected and deployed a responsive front-end interface for a custom academic utility, optimizing campus routing and workload distribution through seamless Quest schedule integration
- Synchronized front-end components with complex backend geographic pathfinding logic (Dijkstra's algorithm) and crowdsourced GraphQL metrics to deliver intuitive data visualizations
- Transformed raw architectural data and dynamic algorithmic outputs into a polished interactive dashboard, bridging heavy backend logic with a user-centric experience

Official Student Ambassador

Apr. 2026 – Present

University of Waterloo, Undergraduate Admissions

Waterloo, ON

- Spearhead digital community engagement initiatives in strategic partnership with UW Admissions, cultivating an inclusive onboarding ecosystem for the incoming undergraduate cohort
- Deliver tailored mentorship and strategic advising to prospective students, navigating complex inquiries regarding academic trajectories, resource allocation, and campus integration
- Execute targeted communication strategies to demystify the university experience, effectively mitigating onboarding friction and driving community retention for hundreds of families

Research Assistant (Young Eagle Project)

May 2023 – May 2024

Chongqing University

Chongqing, China

- Engineered robust data processing workflows to evaluate electricity market trading platforms, leveraging quantitative tools to optimize regulatory pricing models
- Aggregated, synthesized, and visualized high-dimensional project data, delivering actionable insights that significantly accelerated the core research pipeline for senior academic staff

ADDITIONAL PROJECTS & COMPETITIONS

FitFuel (HCI Project) | *Independent Developer*

Jan. 2024 – Apr. 2024

- Conceptualized and prototyped a high-fidelity interactive application utilizing Human-Centered Design (HCD) principles under the mentorship of Dartmouth and Chongqing University faculty
- Designed seamless user flows and wireframes within Figma to directly address user pain points in biometric data management and fitness tracking

International Space Settlement Design Competition | *Winning Company Rep.*

July 2024

- Directed 3D architectural rendering and spatial calculation for aerospace residential structures using advanced CAD environments during a rigorous 42-hour hackathon in Florida, USA
- Partnered within a cross-functional international team to synthesize complex technical specifications into a winning Request for Proposal (RFP) presentation, successfully securing the championship